2.13 Incompressible flow

We can derive a set of equations for incompressible flow that includes:

- mass conservation, Eq. (2.8);
- momentum conservation, Eq. (2.19);
- the material derivative for $\mathbf{u}$, Eq. (2.26);
- the rate of deformation tensor, Eq. (2.33);
- the Newtonian fluid model, Eq. (2.41).

Combining Eq. (2.41) and Eq. (2.33) and using Eq. (2.36) for the deviatoric part of a tensor, gives an expression for stress:

$$\boldsymbol{\sigma} = \mu\left[\nabla\mathbf{u} + \nabla\mathbf{u}^{\mathrm{T}} - \frac{2}{3}\mathrm{tr}(\nabla\mathbf{u})\mathbf{I}\right] - p\mathbf{I}. \tag{2.44}$$

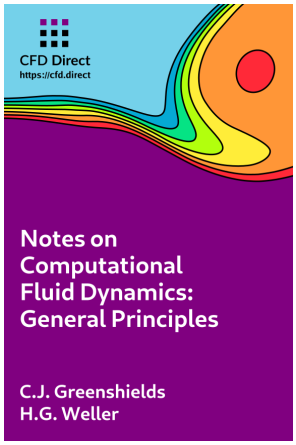
Substituting $\boldsymbol{\sigma}$ into Eq. (2.19) and applying Eq. (2.26) gives an equation for momentum for a Newtonian fluid:

$$\frac{\partial\rho\mathbf{u}}{\partial t} + \nabla\cdot(\rho\mathbf{u}\mathbf{u}) - \nabla\cdot(\mu\nabla\mathbf{u}) - \nabla\cdot\left[\mu(\nabla\mathbf{u})^{\mathrm{T}}\right] + \frac{2}{3}\mu\nabla(\nabla\cdot\mathbf{u}) = -\nabla p + \rho\mathbf{b}. \tag{2.45}$$

The derivation of the equation above uses Eq. (2.35) and the identity $\nabla\cdot(\boldsymbol{\Psi}\mathbf{I}) = \nabla\boldsymbol{\Psi}$ on page 84.

An incompressible fluid exhibits $\rho =$ constant over time, *i.e.* for *moving volumes of fluid* $D\rho/Dt = 0$. Combining the material derivative Eq. (2.14) and mass conservation Eq. (2.8) gives $D\rho/Dt + \rho(\nabla\cdot\mathbf{u}) = 0$, which results in the *incompressibility condition*

$$\nabla\cdot\mathbf{u} = 0. \tag{2.46}$$



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A *homogeneous*, incompressible material exhibits ρ = constant uniformly throughout the *entire fluid*. With that assumption, Eq. (2.45) can be written as

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \bullet (\mathbf{u}\mathbf{u}) - \nabla \bullet (\nu \nabla \mathbf{u}) - (\nabla \nu) \bullet (\nabla \mathbf{u}) = -\nabla p + \mathbf{b}. \tag{2.47}$$

This is the momentum equation for a homogeneous, incompressible Newtonian fluid. It includes: *kinematic* viscosity $\nu = \mu/\rho$ in SI units of $\mathrm{m}^2\mathrm{s}^{-1}$; and p represents *kinematic* pressure, *i.e.* divided by ρ, in SI units of $\mathrm{m}^2\mathrm{s}^{-2}$.

The identity $\nabla \bullet \left[\nu(\nabla \mathbf{u})^{\mathrm{T}}\right] \equiv \nu \nabla(\nabla \bullet \mathbf{u}) + (\nabla \nu) \bullet (\nabla \mathbf{u})$ and the incompressibility Eq. (2.46) yield the terms in Eq. (2.47).

Pressure equation

Mass and momentum conservation, represented by Eq. (2.46) and Eq. (2.47) respectively, provide two equations — one scalar, one vector — for two fields, p and $\mathbf{u}$. However, Eq. (2.46) cannot be solved in its own right since it provides only one equation for vector $\mathbf{u}$, containing 3 components.

A scalar equation, including both p and $\mathbf{u}$, can be derived by taking *the divergence* of Eq. (2.47) and eliminating terms by substituting Eq. (2.46), noting that $\nabla \bullet (\nabla \bullet \nabla \mathbf{u}) \equiv \nabla^2 (\nabla \bullet \mathbf{u})$. For ν and $\mathbf{b}$ that are constant and uniform, the equation is

$$\nabla^2 p + \nabla \bullet [\nabla \bullet (\mathbf{u}\mathbf{u})] = 0. \tag{2.48}$$

This pressure equation can replace Eq. (2.46) to provide a pair of equations, with Eq. (2.47), for both p and $\mathbf{u}$. Chapter 5 describes the algorithms used in the finite volume method which couple the two equations using a modified form of Eq. (2.48).

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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